

Design and Implementation of an Accessible Multimodal Snake and Ladder

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Abstract—This paper presents the design and implementation of an accessible multimodal digital board game based on the classic Snake-and-Ladder game. The system was designed to support inclusive interaction for users with different abilities, particularly visually impaired and motor-impaired users. The application integrates multiple interaction modalities, including voice commands and touch interaction, text-to-speech narration, visual feedback, and haptic feedback.

The main contribution of the project lies in the implementation of a multimodal interaction framework capable of combining heterogeneous inputs and outputs into a coherent interaction experience. The system follows an accessibility-first approach and demonstrates how multimodal fusion can improve usability, flexibility, and interaction redundancy in digital environments.

Index Terms—Multimodal interaction, accessibility, inclusive design, human-computer interaction, assistive technology, voice user interfaces, text-to-speech, multimodal fusion, digital games, universal design.



1 INTRODUCTION

Human-Computer Interaction (HCI) systems traditionally rely on visual graphical interfaces controlled through mouse and keyboard input. Although these paradigms are effective for most users, they may introduce accessibility barriers for people with visual impairments, motor disabilities, or interaction constraints.

Multimodal interaction aims to address these limitations by allowing users to interact with systems through multiple complementary modalities such as speech, touch, keyboard input, visual feedback, and audio output. Integration of multiple channels improves flexibility, robustness, and accessibility, while reducing dependence on a single interaction mechanism.

This project explores these concepts through the implementation of an accessible digital Snake-and-Ladder game designed for Android devices. The primary objective is not the complexity of the game mechanics themselves, but rather the design and implementation of an inclusive multimodal interaction system capable of supporting different user abilities and preferences.

The system integrates voice commands, speech synthesis, visual feedback, and haptic feedback into a unified interaction framework. In addition, the project adopts accessibility-oriented design principles to ensure that all main game functionalities can be accessed through multiple modalities.

2 RELATED WORK

Multimodal interaction has been extensively studied within the field of Human-Computer Interaction. Early multimodal systems demonstrated that the combination of heterogeneous input modalities improves interaction robustness and reduces user effort. Oviatt highlighted how multimodal interfaces can naturally support redundancy and

error correction by allowing users to switch dynamically between modalities.

Accessibility research also emphasizes the importance of alternative interaction channels for users with disabilities. Voice User Interfaces (VUIs), text-to-speech systems, and haptic feedback technologies have become fundamental components of assistive interaction systems. The concept of Universal Design advocates for interfaces that are accessible to the widest possible audience without requiring adaptation.

Digital games have increasingly been used as environments for accessibility experimentation due to their interactive nature and clear feedback mechanisms. However, many existing games still rely heavily on visual interfaces and precise motor interaction, which limit accessibility for certain categories of users.

The proposed system combines multimodal interaction principles and accessibility-oriented design within a simple board game environment, demonstrating how inclusive interaction can be integrated into entertainment applications.

3 SYSTEM OVERVIEW

The developed application is a digital implementation of the classic Snake-and-Ladder board game. The game supports two players and reproduces the standard mechanics of dice rolling, board movement, snakes, ladders, and turn-based gameplay.

The architecture of the system was designed to separate game logic from interaction management in order to improve modularity and extensibility. The application is composed of four main subsystems:

- Input Management System
- Multimodal Fusion Module
- Game Logic Engine
- Output and Feedback System

The Input Management System handles heterogeneous interaction modalities, including voice commands, touch interaction, and accessible buttons. The Multimodal Fusion Module interprets and maps user inputs into semantic game actions. The Game Logic Engine updates the game state and coordinates movement and gameplay events. Finally, the Output System synchronizes visual, auditory, and haptic feedback.

The application was developed for Android using Kotlin and Jetpack Compose.

4 MULTIMODAL INTERACTION DESIGN

4.1 Input Modalities

The system supports multiple input modalities to improve accessibility and interaction flexibility.

The voice interaction is implemented through the Android SpeechRecognizer API. Users can issue spoken commands such as “roll dice”, “restart”, or “back to setup”. The system processes speech recognition results and maps semantically similar utterances to common game actions. This approach improves robustness and reduces the rigidity of command recognition.

Traditional touch interaction is also supported through accessible graphical buttons implemented using Jetpack Compose. The buttons were specifically designed to support accessibility-oriented interaction patterns. Rather than immediately activating actions, the interface first announces the selected action through speech synthesis and requires confirmation through a second interaction. This behavior reduces accidental activations and improves usability for visually impaired users.

The system architecture allows users to dynamically switch between modalities depending on contextual needs or personal preferences.

4.2 Output Modalities

The application provides multiple coordinated output modalities in order to ensure accessibility redundancy and improve interaction clarity.

Visual feedback is provided through a graphical board interface showing player positions, dice values, and current game status. The interface uses large components, high-contrast colors, and simplified layouts to improve readability.

Auditory feedback is implemented through the Android TextToSpeech engine. The system narrates game events such as dice rolls, player movements, turn changes, ladder climbing, snake descent, and victory conditions. This functionality allows visually impaired users to interact with the game without relying exclusively on visual information.

The application also integrates haptic feedback through vibration patterns. Different vibration sequences are associated with specific events such as movement, ladder climbing, snake descent, and victory notifications. Haptic feedback provides an additional non-visual communication channel and reinforces multimodal accessibility.

4.3 Multimodal Fusion

A central aspect of the project is the implementation of a multimodal fusion strategy capable of integrating heterogeneous inputs into a unified interaction framework.

The system follows a late fusion approach in which each modality is processed independently before being mapped into a shared semantic action space. Voice commands, touch events, and accessible button interactions are interpreted separately and subsequently converted into abstract game actions such as rolling dice, restarting the game, or navigating between screens.

This design improves robustness and interaction flexibility. For example, the same game action can be triggered through a spoken command or through a graphical button. This redundancy allows users to select the interaction modality that best matches their abilities and contextual constraints.

The fusion mechanism also coordinates asynchronous events generated by different modalities. Speech recognition, animations, speech synthesis, and vibration feedback are synchronized to maintain consistent interaction behavior and coherent user feedback.

5 ACCESSIBILITY CONSIDERATIONS

Accessibility was considered a primary design objective throughout the entire development process.

For visually impaired users, the system minimizes dependency on purely visual interaction through extensive use of text-to-speech narration and audio feedback. Important events are always communicated through speech synthesis, allowing users to understand the game state without continuously observing the screen.

The interface design also includes accessibility-oriented graphical choices such as high-contrast colors, large buttons, simplified layouts, and reduced visual clutter.

For users with motor impairments, the system reduces reliance on precise touch interaction by supporting voice commands for all major actions. In addition, the accessible button interaction model minimizes accidental activations by introducing explicit confirmation mechanisms.

The project also considers cognitive accessibility by maintaining simple game mechanics, consistent feedback patterns, and a predictable turn-based structure that reduces cognitive load.

Rather than treating accessibility as an optional feature, the system was designed around accessibility-first principles from the earliest development stages.

6 SYSTEM ARCHITECTURE

The software architecture follows a modular event-driven structure designed to support multimodal coordination.

The interaction pipeline can be summarized as follows:

User Input → Input Manager → Fusion Module → Game Engine → Output Manager

The Input Manager receives interaction events from multiple modalities, including speech recognition and touch interaction. The Fusion Module interprets these inputs and maps them into semantic commands understood by the

Game Engine. The Game Engine updates the application state and generates output events that are distributed across graphical, auditory, and haptic feedback channels.

The modularity of the architecture improves maintainability and allows future integration of additional modalities such as gesture recognition or eye-tracking systems.

7 IMPLEMENTATION TECHNOLOGIES

The application was implemented using modern Android development technologies.

The graphical interface was developed using Jetpack Compose, which simplifies declarative UI construction and improves state synchronization. Kotlin was used as the primary programming language for implementing both application logic and interaction management.

Speech recognition functionality was implemented through the Android SpeechRecognizer API, while speech synthesis relied on the TextToSpeech API. Haptic feedback was implemented through the Android vibration system using custom vibration patterns associated with different gameplay events.

The modularity of the Android framework facilitated the integration of multiple synchronized modalities within a single interaction environment.

8 TECHNICAL CHALLENGES

One of the main challenges encountered during development involved the synchronization between asynchronous multimodal events. Speech recognition, speech synthesis, movement animations, and haptic feedback operate asynchronously and required careful coordination to avoid inconsistent interaction states.

Another significant challenge concerned the reliability of speech recognition. Variations in pronunciation, ambient noise, and microphone quality affected command recognition accuracy. To mitigate this issue, the system supports multiple semantically equivalent commands for the same action.

Managing the flow of interaction during animations also required additional synchronization logic. During token movement animations, certain actions are temporarily disabled to prevent inconsistent game states.

Finally, balancing accessibility and simplicity was an important design consideration throughout the project.

9 EVALUATION

The application was evaluated through informal testing sessions that include different interaction scenarios, including voice-only interaction, touch-only interaction, and combined multimodal usage.

The evaluation demonstrated that multimodal interaction significantly improves interaction flexibility and accessibility. Voice commands effectively reduced the reliance on precise touch interaction, while text-to-speech narration improved usability for visually constrained interaction scenarios.

Haptic feedback was found to be particularly useful in reinforcing important gameplay events and improving the immediacy of the feedback.

However, several limitations emerged during the testing. Speech recognition performance decreased in noisy environments, and continuous voice interaction introduced occasional recognition delays. In addition, the current system does not yet support adaptive mode prioritization based on user behavior or environmental context.

Despite these limitations, the overall results suggest that multimodal fusion can significantly improve accessibility and robustness of interaction in digital games.

10 LIMITATIONS AND FUTURE WORK

Although the developed system successfully demonstrates accessible multimodal interaction principles, several limitations remain.

The current implementation supports only a limited set of predefined voice commands and relies on keyword-based semantic mapping rather than advanced natural language understanding. Furthermore, the application currently supports only local gameplay and does not include adaptive personalization mechanisms.

Future work may involve integrating machine learning techniques for intent recognition and adaptive mode management. Additional modalities such as gesture recognition, eye tracking, or external assistive devices could also be incorporated.

A formal usability study involving real users with accessibility needs would provide more comprehensive evaluation results and help identify further improvements.

11 CONCLUSION

This paper presented the design and implementation of an accessible multimodal Snake-and-Ladder game aimed at supporting inclusive Human-Computer Interaction.

The system integrates multiple coordinated modalities including speech recognition, touch interaction, text-to-speech narration, graphical feedback, and haptic feedback. Through multimodal fusion, the application provides redundant and complementary interaction pathways that improve accessibility and usability.

The project demonstrates how multimodal interaction principles can be applied effectively within a simple game environment to create more inclusive digital experiences. The results highlight the importance of accessibility-first design approaches and suggest promising directions for future research in multimodal and assistive interaction systems.

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